

host_agent

Initialised

User: Find all the tall houses that are over 18m or 7 storeys tall in Exeter

planning_agent

The area specified is Exeter, so there will be 1 entry for Exeter. The entity to search for is tall houses, defined as those over 18m or 7 storeys tall. The condition is to find all such houses in Exeter.

output steps: ["Find Exeter 1 area", "Search for tall houses in Exeter (over 18m or 7 storeys tall) as many entries as possible"]

named_area_agent

Tool call: call_os_ngd - Arguments: {'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_area_polygon'} - Response: Search results have been found. Artifact generated is : exeter_area_polygon, Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter., count of records fetched : 98.

Tool send_message - Arguments: {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_area_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'."}

data_analysis_agent

Tool generate_metadata_for_artifacts - Arguments: {'artifact_names': ['exeter_area_polygon']}

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
```

```
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for Exeter using multiple relevant columns for high quality
    filtered = gdf[
        (
            (gdf['name1_text'].str.lower() == 'exeter') |
            (gdf['name1_text'].str.lower().str.contains('exeter'))
        ) & (
            (gdf['descriptiongroup'].str.lower().isin(['settlement', 'part of settlement'])) |
            (gdf['description'].str.lower().isin(['city', 'town', 'suburban area']))
        )
    ]
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter using name and description columns."
    return [
        summary,
        'filtered_exeter_area_polygon',
        'GeoDataFrame containing only the polygon(s) representing Exeter, filtered using name1_text, descriptiongroup, and description columns for high accuracy',
        filtered
    ]
```

- Output: Filtered to 1 polygon(s) for Exeter using name and description columns.

Artifact name: filtered_exeter_area_polygon Description: GeoDataFrame containing only the polygon(s) representing Exeter, filtered using name1_text, descriptiongroup, and description columns for high accuracy.

named_area_agent

The specific polygon for Exeter has been filtered and saved as:

- **Artifact Name:** filtered_exeter_area_polygon
- **Description:** GeoDataFrame containing only the polygon(s) representing Exeter, filtered using name and description columns for high accuracy.

buildings_agent

Tool call: call_os_ngd - Arguments: {'filters': None, 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': None, 'filename': 'all_buildings_in_filtered_exeter_area'} - Response: Multiple search results have been found in multiple datasets. A summary of each is provided. Artifacts generated are : ['buildings_all_buildings_in_filtered_exeter_area', 'buildingline_all_buildings_in_filtered_exeter_area',

'buildingpart_all_buildings_in_filtered_exeter_area'], Descriptions are : ["A geopandas dataframe containing building data with filters and bbox applied for the query Search for all houses in the area defined by 'filtered_exeter_area_polygon' that are over 18m tall or 7 storeys tall. using filters None.", ...], counts of records fetched are : [69169, 69342, 85480].

Tool send_message - Arguments: {'target': 'data_analysis_agent', 'task_description': "From the artifact 'buildings_all_buildings_in_filtered_exeter_area', filter for houses, defined as buildings where buildinguse_addresscount_residential > 0 and buildinguse_addresscount_total = 1. Then, from these houses, select those that are either over 18m tall (relativeroofbase > 18) or have 7 or more storeys (storeycount >= 7). Save the filtered results as 'houses_over_18m_or_7storeys_filtered_exeter_area'."}

data_analysis_agent

Tool generate_metadata_for_artifacts - Arguments: {'artifact_names': ['buildings_all_buildings_in_filtered_exeter_area']}

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_houses_over_18m_or_7storeys(data: list):
    import geopandas as gpd
    df = data[0]
    # Step 1: Filter for houses: residential > 0 and total = 1
    houses = df[(df['buildinguse_addresscount_residential'] > 0) & (df['buildinguse_addresscount_total'] == 1)]
    # Step 2: Filter for height or storeys
    filtered = houses[(houses['height_relativeroofbase_m'] > 18) | (houses['numberoffloors'] >= 7)]
    summary = f"Filtered houses in Exeter area with either height > 18m or 7+ storeys. Total: {len(filtered)} entries."
    artifact_name = 'houses_over_18m_or_7storeys_filtered_exeter_area'
    artifact_description = 'GeoDataFrame of houses (residential, single address) in Exeter area that are either over 18m tall (relativeroofbase > 18) or have 7 or more storeys'
    return [summary, artifact_name, artifact_description, filtered]
```

- Output: The search return no results and no artifacts are generated

No houses in the filtered Exeter area meet the criteria of being both residential (with a single address) and either over 18m tall or having 7 or more storeys. No results were found for these conditions.

buildings_agent

No houses in the area defined by 'filtered_exeter_area_polygon' meet the criteria of being both residential (with a single address, as per the definition of a house) and either over 18 meters tall or having 7 or more storeys. No results were found for these conditions.

If you would like to adjust the criteria or search for a different type of building, please let me know!

host_agent

No houses in Exeter meet the criteria of being over 18 meters tall or having 7 or more storeys, according to the available building data. If you would like to adjust your criteria or search for a different type of building, please let me know!